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A Mini Project Report

On

“BOREWELL SAFETY ALERT SYSTEM FOR CHILDREN ”

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Electronics & Communication Engineering

Submitted by

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DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

(B.E (E&C) Program Accredited by NBA, New Delhi, Validity 01.07.2017 to 30.06.2020 & 01.07.2020 – 30.06.2023 & 01.07.2023 to 30.06.2026)

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CERTIFICATE

Certified that the 5th Semester Mini-Project titled “ **Borewell Safety Alert System for Children**” is a bonafide work carried out by **Khushi A(4GW23EC067)**, **Manya M (4GW23EC085)**, **Nidhi A Togari(4GW23EC099)** and **Priyanka N(4GW23EC119)** in partial fulfillment for the award of degree of Bachelor of Engineering in **Electronics & Communication Engineering** of the Visvesvaraya Technological University, Belagavi, during the year 2025-26. The Mini-Project report has been approved as it satisfies the academic requirements with respect to the project work prescribed for Bachelor of Engineering Degree.

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ABBREVIATIONS

IoT	Internet of Things
LCD	Liquid Crystal Display
GSM	Global System for Mobile Communication
GPS	Global Positioning System
PLC	PROGRAMMABLE LOGIC CONTROLLER
SCADA	Supervisory Control And Data Acquisition

CHAPTER 1

INTRODUCTION

In recent years, the number of tragic incidents involving children accidentally falling into uncovered or abandoned borewells has significantly increased. Borewells, which are narrow and deep structures used for extracting groundwater, often remain unsealed or inadequately covered after their purpose is served. Due to their small diameter and great depth, rescuing a trapped child becomes extremely difficult, time-consuming, and sometimes unsuccessful, resulting in severe injuries or loss of life.

To address this critical safety concern, a Borewell Safety Alert System is essential. This system aims to detect unauthorized or accidental entry near the borewell and immediately alert nearby individuals or authorities. By integrating sensors, alarms, and communication modules, the system enhances safety and enables faster response to prevent fatal accidents.

The primary objective of this project is to design and develop a reliable safety mechanism that can continuously monitor the borewell area and provide alerts in real time. Implementing such preventive technology can play a vital role in safeguarding young children and reducing borewell-related incidents across rural and urban areas.

1.1 Problem Statement

Borewell accidents involving children have become a recurring tragedy in many regions, with numerous cases of children falling into uncovered or poorly secured borewells. Traditional safety measures like physical covers are often inadequate, easily removed, or forgotten after maintenance. The absence of proactive warning systems and emergency response mechanisms creates life-threatening situations. Parents and guardians lack effective tools to monitor children's proximity to these hazardous sites, and emergency response is often delayed due to late discovery of accidents. There is a critical need for an intelligent safety system that can proactively prevent children from approaching borewells, provide immediate alerts when boundaries are breached, and enable rapid emergency response with precise location information. This project aims to develop a comprehensive borewell safety solution that combines geo-fencing technology with emergency communication to protect children and provide peace of mind to parents and communities.

1.2 Aim

The aim of this project is to develop a smart Borewell Safety Alert System using sensors and communication technology to monitor borewell sites, detect accidental falls of children, and immediately alert nearby authorities or guardians to ensure rapid rescue and prevent fatalities.

1.3 Objective

- To develop a borewell safety system using Node MCU with GPS and GSM integration
- To implement geo-fencing technology for creating virtual safety boundaries, create a panic button also implement audible alerts using buzzer , communication via SIM800L with SMS alerts

1.4 Scope of the project

The scope of this project is to design and develop an efficient and intelligent Borewell Safety Alert System that can actively prevent accidents involving children near open or abandoned borewells. The system focuses on creating a protective virtual boundary using geo-fencing technology, which continuously tracks movement around the danger zone and immediately detects when a child or any person enters the restricted area. The project includes the integration of NodeMCU, GPS, GSM, and a panic alert mechanism to ensure continuous monitoring, accurate location tracking, and automatic emergency notifications to parents, nearby residents, and authorities. The system functions as an early-warning preventive model rather than a rescue-based solution, helping to reduce response time and avoid tragic incidents. The project further involves developing a low-power, cost-effective prototype that is suitable for rural areas where borewell openings are often left unattended. It aims to provide real-time communication through SMS alerts and maintain reliable connectivity even in remote locations. The system also explores the possibility of adding smart wearables for children that work together with the main unit to strengthen safety.

1.5 Organization of the report

This project report is organized into six chapters, each detailing a specific stage of development of the Borewell Safety Alert System for Children:

- **Chapter 1: Introduction**

This chapter provides the background of the problem, the significance of borewell safety, and a brief literature review. It also includes the problem statement, aim, objectives, methodology overview, scope of the project, and limitations.

- **Chapter 2: Literature Survey**

This chapter presents various studies, existing technologies, and previous approaches related to borewell safety solutions, highlighting the research gap that motivates the proposed system.

- **Chapter 3: Methodology**

This chapter explains the working principles, design approaches, hardware and software integration, and overall system framework used to develop the proposed solution.

- **Chapter 4: Implementation and Flow Chart**

This chapter describes the step-by-step implementation, components used, system connections, flow charts, and block diagrams that illustrate the operational process.

- **Chapter 5: Results and Discussions**

This chapter discusses the experimental results, system performance evaluation, and observations. Graphs, output images, and analysis are included to validate the effectiveness of the system.

- **Chapter 6: Conclusions and Future Scope**

This chapter summarizes the work carried out, highlights the achievements of the proposed system, and suggests possible improvements and future enhancements for real-world implementation.

CHAPTER 2

LITERATURE SURVEY

Literature review is the survey done on the previous research and the projects carried out. This chapter includes review on 05 different papers published at various journals and conferences. It provides us with a greater knowledge and wider range of exposure in the area of the topic being dealt.

2.1 LITERATURE SURVEY

[1]V. Priyanka, B. P. Jeyanthi, and S. Vasanthakumar, **“Intelligent Borewell Safety System Using Proximity Sensors and Automated Closure Mechanism,”** Journal of Artificial Intelligence and Computer Science, vol. 5, no. 1, pp. 12–18, 2025. The authors developed a proximity sensor-based system capable of detecting human presence near borewell openings. A relay automatically triggered the closure of the borewell lid, reducing the probability of unnoticed falls. The system lacked GSM/GPS modules, limiting remote alert capabilities. Future work suggested geofencing and advanced alert notifications.

[2]Ramya D., Dilip S., Deepiga M., and Gokulanatha B., **“Borewell rescue system based on Node MCU (ESP8266), acoustic sensors, conductivity sensors, and GSM modules”** published in the International Research Journal of Modernization in Engineering and Technology Science (2025), proposed a. Their system effectively detected trapped children in open borewells by monitoring sound, water levels, and movement while generating automated alarms. The prototype demonstrated quick local alerts and real-time sensor processing. However, the authors noted that the design lacked wearable child-tracking devices and geofencing capabilities, suggesting further improvement by integrating GPS-based safety measures.

[3]A. M. Ruben et al., “**Smart Borewell Child Rescue System Using Geofencing**”, International Journal of Engineering Research and Technology, vol. 12, no. 4, pp. 77–82, 2024. The study introduced a system capable of real-time hazard detection and child tracking inside borewells. By integrating Node MCU, GPS, and geofencing, the system could generate emergency alerts and support robotic retrieval mechanisms. Despite its effectiveness, the authors recommended AI integration and cloud-based analytics for improved decision-making.

[4]D. Mounika and D. Deepika, “**Multi-Sensor Borewell Rescue System Using IoT Nodes and Visual Monitoring**,” Semantic Scholar, 2023. This system used ESP8266, temperature sensors, MQ135, DHT11, and camera modules for borewell condition analysis. It provided real-time environmental monitoring including gas levels and temperature changes, along with visual feedback. The authors recommended adding GSM/GPS modules for remote alerts and improved guardian notifications

[5]S. Bharathi, “**AI-Based Smart Borewell Rescue System Using Camera Modules**,” JES Publication, vol. 8, no. 3, pp. 23–29, 2023. The proposed concept employed camera modules and AI-based processing for quick identification of trapped children and distress detection. The authors emphasized the need for integrating low-power GSM modules, GPS units, and wearable panic buttons to improve field deployment and rescue coordination.

[6]S. Suresh Kumar et al., “**Wireless Child Monitoring System for Borewells Using AI and IoT**,” Journal of Novel Research and Development, vol. 2, no. 6, pp. 112–119, 2023. This work used AI and IoT for detecting the presence and condition of children trapped inside borewells. The system monitored sensor data in real time and generated alerts through GSM. Although the system performed well, the authors suggested improvements including lightweight ESP modules, GPS tagging, and integration with government emergency networks.

[7]S. Shreya, M. Kumar, and R. Deepak, **“GSM-Enabled Borewell Safety Monitoring System Using Ultrasonic and PIR Sensors,”** International Journal of Engineering Research & Technology, vol. 13, no. 4, pp. 55–59, 2024. The authors developed a safety system that employed ultrasonic and PIR sensors to detect human presence and depth variations near borewell openings. Alerts were transmitted using GSM to stakeholders. Although the system effectively minimized unnoticed falls, it lacked camera-based verification, which the authors suggested as a future enhancement

[8]R. S. Kumar and M. Priyadharshini, **“Smart Borewell Risk Prevention Using ESP8266 and Proximity Sensors,”** International Journal for Research in Applied Science and Engineering Technology, vol. 11, no. 7, pp. 221–226, 2023. This work utilized ESP8266, proximity sensors, and soil-moisture detection for monitoring unsafe conditions around borewells. GSM was used for alert transmission. While efficient in hazard identification, the system lacked GPS-based coordinate tracking, limiting the precision of emergency response.

[9]M. Ramesh, S. Prasad, and K. Anitha, **“AI-Assisted Child Fall Detection System for Borewell Safety Using Camera Modules,”** Proceedings of ICCTEST 2023, pp. 310–315, 2023. The authors proposed a camera-based surveillance method supported by lightweight CNN models to detect child movement near borewells. Real-time cloud alerts were generated upon detecting risk. The system showed high detection accuracy but required stable network connectivity for effective functioning.

[10]H. D. Rao, M. Naveen, and S. Varshini, **“Automated Borewell Protection Lid Using Proximity Sensors and GSM Alerts,”** International Journal of Electrical and Electronics Engineering, vol. 12, no. 2, pp. 48–53, 2024. The researchers designed an automatic borewell lid using servo motors and proximity sensors. GSM alerts were sent when human movement was detected around the borewell. Though effective in closure automation, the system lacked environmental sensing capabilities, especially gas monitoring

CHAPTER 3

METHODOLOGY

This chapter includes the proposed methodology, which plays a major role in the project. It also describes the hardware requirements and the software requirements of our project

3.1 PROPOSED METHODOLOGY

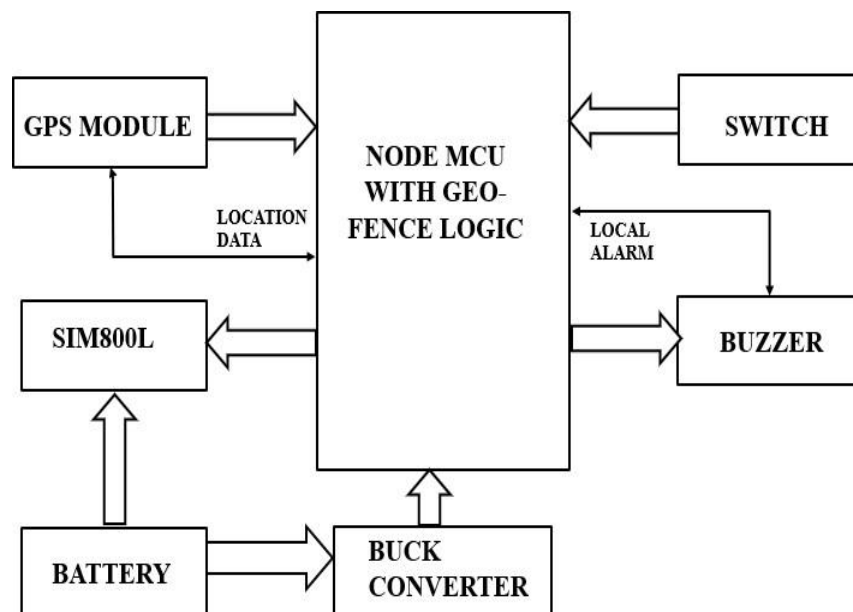


Fig 3.1: Block diagram of Borewell Safety Alert System for Children

The block diagram in fig 3.1 illustrates the working methodology of the Borewell Safety Alert System, where the entire operation is centered around the NodeMCU microcontroller programmed with geo-fencing logic. The GPS module continuously collects the real-time location data of the monitored area and sends this information to the NodeMCU. Based on the received coordinates, the NodeMCU checks whether any movement or object has entered the predefined restricted safety boundary created around the borewell. Whenever the system detects a breach of this virtual boundary, it activates both the communication

and alert mechanisms to immediately warn people nearby and notify responsible authorities.

Children To ensure that emergency messages are delivered instantly, the SIM80 GSM module is connected to the NodeMCU. When the geo-fence is crossed, the NodeMCU sends a command to the GSM module, which then transmits SMS alerts to the registered phone numbers of parents, neighbors, or local officials. This helps in rapid response and prevents accidents by creating immediate awareness. Along with remote alerts, a buzzer is also connected to provide a loud local alarm at the site itself. This buzzer is triggered automatically by the NodeMCU whenever danger is detected, ensuring that anyone nearby gets notified even if they are not carrying a phone or have not received the message.

A switch is added to manually enable or disable the system during installation, maintenance, or battery replacement. The entire setup is powered by a rechargeable battery, and a buck converter is used between the battery and control unit to regulate the voltage and supply a stable power source to all components. This protects sensitive modules such as the NodeMCU, GPS, and SIM800L from voltage fluctuations and ensures smooth, uninterrupted functioning. Overall, the block diagram represents a smart and interconnected safety system in which every component works together to monitor the borewell area, detect danger in advance, and alert people instantly, thereby preventing accidents and enhancing child safety.

3.2 HARDWARE REQUIREMENTS

This subsection gives the detailed information of each of the hardware component used in the system

- Node MCU (ESP8266)
- NEO-6M GPS Module
- SIM800L GSM Module
- Panic Button Switch
- Piezo Buzzer
- Lithium Battery (3.7V or 7.4V)
- Buck Converter Module

3.2.1 Node MCU (ESP8266)

In this project, the NodeMCU ESP8266 fig 3.2 acts as the central intelligent controller that manages the entire borewell safety system. It continuously collects information from the GPS module, checks whether the predefined geo-fence limit is crossed, reads the panic button status and then decides whether to activate the alert mechanism. The NodeMCU processes all sensor data in real time and controls the GSM module and buzzer based on danger conditions. Its builtin Wi- Fi capability also allows optional IoT monitoring if required, making it the brain of the system.



Fig 3.2: Node MCU (ESP8266)

3.2.2 NEO-6M GPS Module

The NEO-6M GPS fig 3.3 module plays a vital role by providing accurate live latitude and longitude values. This GPS information helps the NodeMCU determine the exact position of the child or device around the borewell area. Using GPS distance calculations, the controller creates a virtual boundary or geo-fence. Whenever the GPS module shows movement outside the safe radius, it is immediately recognized as a risk condition. Moreover, the NEO-6M continuously updates its coordinates, allowing the

system to track even slight movements near the borewell. Its high sensitivity ensures reliable data even in rural or low-signal areas.

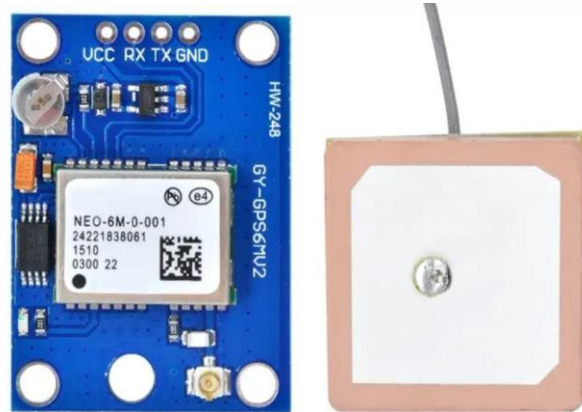


Fig 3.3: NEO-6M GPS Module

3.2.3 SIM800L GSM Module

The SIM800L GSM module fig 3.4 is responsible for sending emergency alerts to parents, nearby residents and authorities. When the NodeMCU detects that the geo-fence is crossed or when the panic button is pressed, it sends AT commands to the GSM module. The module then transmits an SMS containing the danger message along with GPS coordinates. This ensures that help can reach the location quickly even if no one is physically present near the borewell at that moment, making the system reliable for remote and rural areas.



Fig 3.4: SIM800L GSM Module

3.2.4 Panic Button Switch

The panic button fig 3.5 provides a manual emergency trigger. If the child or any person nearby feels unsafe or notices someone approaching the borewell, pressing the button instantly activates the alert system. This allows the system to send emergency messages even before the GPS boundary is crossed. It strengthens safety by enabling human-initiated alerts during sudden or unexpected situations.



Fig 3.5: Panic Button Switch

3.2.5 Piezo Buzzer

The piezo buzzer fig 3.6 is used to produce a loud alarm sound whenever the system detects danger. As soon as the NodeMCU identifies risk, it activates the buzzer. The sound warns people nearby so they can rush immediately to stop the child from going closer to the borewell. This real-time audible alert is crucial because it provides immediate awareness on the spot.



Fig 3.6: Piezo Buzzer

3.2.6 Lithium Battery (3.7V or 7.4V)

To power the entire unit, a lithium battery fig 3.7 of 3.7V or 7.4V is used. This battery makes the device portable and suitable for outdoor borewell environments where regular power supply may not be available. Since the modules require stable and specific voltages for safe operation, a buck converter is included in the circuit. The buck converter steps down the battery voltage to a regulated 5V or 3.3V as required by the NodeMCU, GPS and GSM modules. It protects the components from over-voltage and ensures reliable long-term operation.



Fig 3.7: Lithium Battery (3.7V or 7.4V)

3.2.7 Buck Converter Module

In the Borewell Safety Alert System, the buck converter fig 3.8 plays a crucial role by ensuring that all electronic components receive the correct and stable operating voltage required for reliable functioning. Since the system is powered using a 3.7V or 7.4V lithium battery, the raw battery output is not suitable for directly powering sensitive modules like the NodeMCU ESP8266, NEO-6M GPS and the SIM800L GSM module, which require regulated voltage to avoid malfunction or permanent damage. The buck converter steps down and stabilizes the higher battery voltage to a safe and constant 5V or 3.3V output depending on the needs of each module. This efficient DC-DC conversion ensures long battery life and prevents voltage fluctuations caused by GSM transmission spikes or continuous GPS operation. By delivering clean and regulated power, the buck converter guarantees the uninterrupted performance of the controller, communication modules and

alarm system, making it an essential power- management component for the entire borewell safety device.

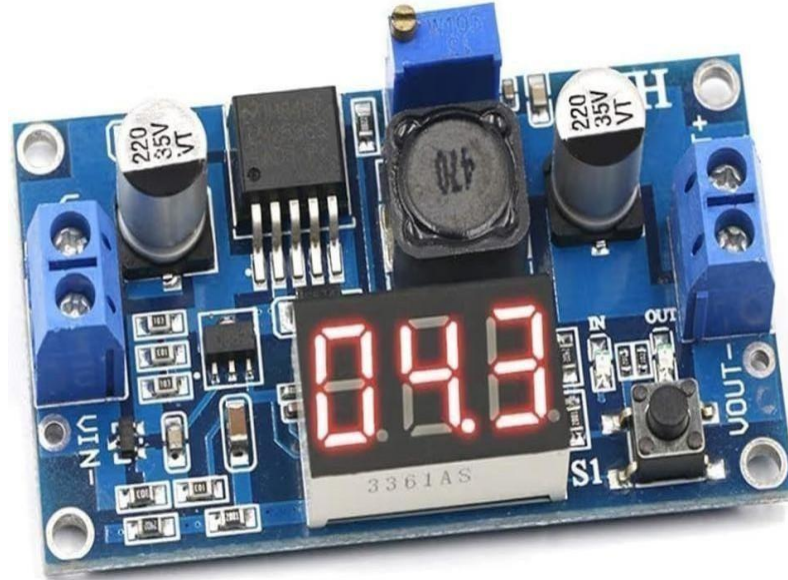


Fig 3.8: Buck Converter Module

3.3 SOFTWARE REQUIREMENTS

This subsection provides detailed information about the software tools and platforms used for the development, programming, testing, and simulation of the Borewell Safety Alert System. The following software resources are utilized in the system:

- Arduino IDE
- Geo-fencing algorithms

3.3.1 Arduino IDE

Arduino is an open-source electronics platform widely used for building interactive and automated systems. It consists of easy-to-use hardware boards and a simple programming environment, making it suitable for beginners as well as advanced developers. Arduino boards are built around microcontrollers that can read inputs from various sensors, process the data, and control outputs such as motors, buzzers, or communication modules. The Arduino IDE provides a user-friendly interface where programs, known as sketches, are

written in a simplified version of C/C++. One of the major strengths of Arduino is its extensive library support, which allows rapid integration of modules like GPS, GSM, Wi-Fi, Bluetooth, and many types of sensors. Because of its flexibility, low cost, and large global community, Arduino has become a popular platform for education, prototyping, and real-world applications across fields such as automation, robotics, environmental monitoring, and IoT.



3.3.2 Geo-fencing algorithms

Step 1: Initialize System:

Start the microcontroller and initialize the GPS module.

Step 2: Set Geo-Fence Boundary:

Store the latitude and longitude of the restricted/safe location as the reference point.

Step 3: Read GPS Coordinates Continuously:

The GPS module provides continuous updates of the current latitude and longitude.

Step 4: Check GPS Validity:

Ensure the received coordinates are valid before processing.

Step 5:Compare With Radius:

If the calculated distance is greater than the predefined radius(15 meter), it means the boundary is crossed.

Step 6:Trigger Alert:

When the system detects a boundary violation, it activates the buzzer and sends an SMS alert with the live location link.

Step 7:Reset or Continue Monitoring:

After the alert, the system continues monitoring for further movements.

CHAPTER 4

IMPLEMENTATION AND FLOW CHART

4.1 Implementation

The implementation of the Borewell Safety Alert System involves both hardware integration and software programming to achieve effective child safety monitoring. The system is developed using NodeMCU as the central controller, which interfaces with GPS, GSM, sensors, and alert mechanisms.

The major steps involved in the implementation are:

4.1.1 Hardware Setup

- NodeMCU ESP8266 is connected to the GPS module to continuously receive latitude and longitude coordinates.
- GSM (SIM800L) module is interfaced to send emergency SMS alerts to the registered mobile number.
- A buzzer is connected to raise an audible alert when a child approaches the borewell area.
- A panic button is included to allow manual emergency triggering.
- The system is powered through a rechargeable battery and regulated using a buck converter.

4.1.2 Software Setup

- The Arduino IDE is used to program the NodeMCU.
- GPS data is processed and compared with predefined geofencing boundaries.
- When a breach or fall is detected:
 - Buzzer rings immediately

4.1.3. System Testing

- The system is tested under different GPS signal conditions.
- GSM network connectivity is verified by sending alerts multiple times.
- Alarm and panic button responses are monitored for reliability.

This integrated approach ensures timely detection and fast communication in emergency situations, making the solution practical for borewell safety.

4.2 Flow Chart

The following flow chart represents the working sequence of the Borewell Safety Alert System:

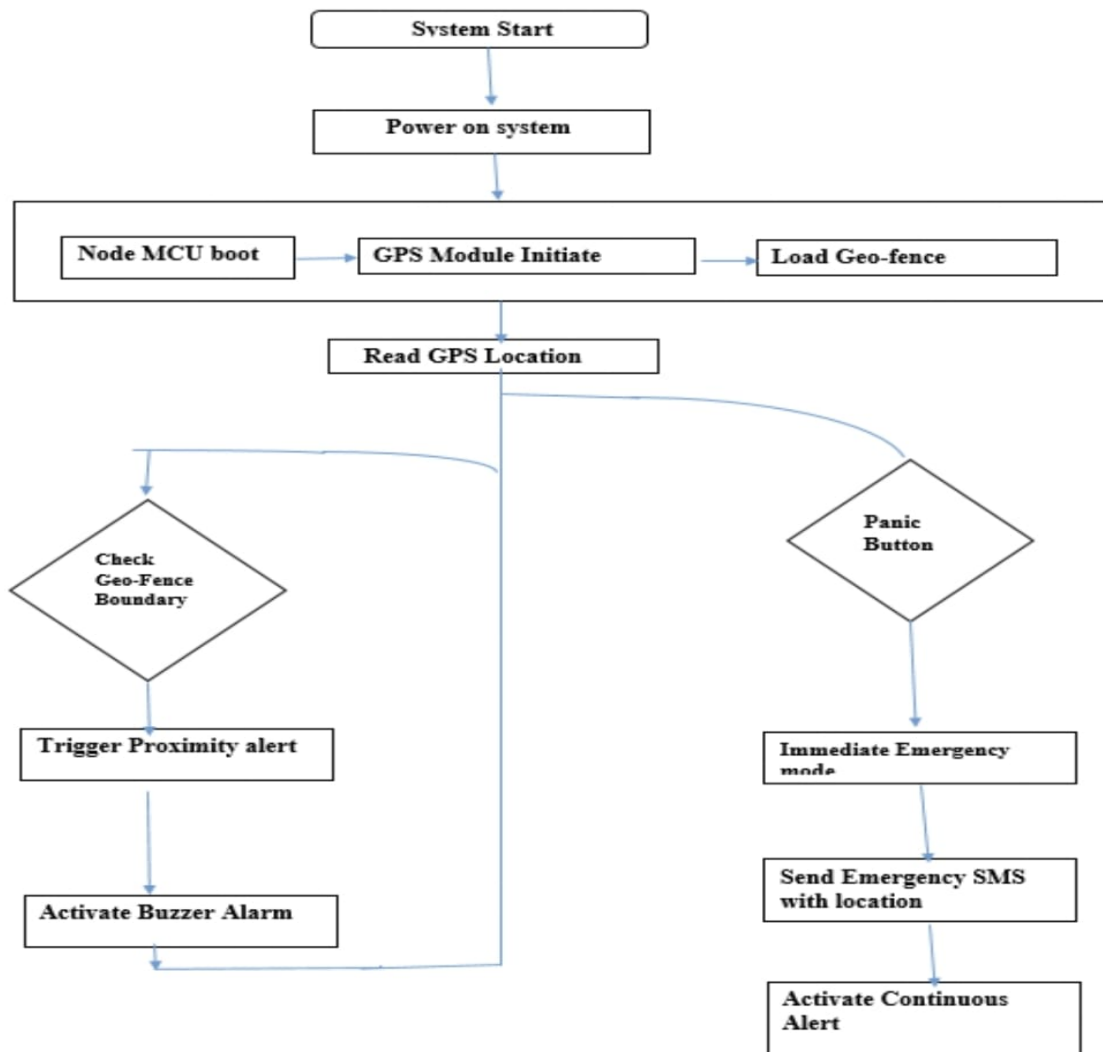


Fig4.3: Flow Chart of Borewell Safety Alert System for Children

The above fig 4.3 illustrates the sequential working of the borewell safety alert system, starting from system initialization to emergency alert activation. It shows how the system continuously monitors location and triggers alerts when a child enters the danger zone.

1. System Start:

The operation begins when the system is turned on.

2. Power On System:

All hardware components receive power and become active.

3. NodeMCU Boot:

The microcontroller initializes and loads the programmed instructions.

4. GPS Module Initialization:

The GPS module starts acquiring satellite signals to provide accurate latitude and longitude values.

5. GSM Module Initialization:

The GSM module prepares for message transmission and communication functions.

6. Load Geo-Fence Coordinates:

The predefined latitude and longitude of the restricted area are loaded into the system.

7. Read GPS Location:

The system continuously reads the real-time GPS coordinates of the user or monitored object.

8. Check Geo-Fence Boundary:

The current location is compared with the geo-fence radius to determine if it lies inside or outside the safe zone.

9. Safe Distance Condition:

If the user is within the safe boundary, the system continues monitoring without triggering any alert.

10. Child/Person Approaching Danger:

If the user moves beyond the defined geo-fence radius, the system identifies entry into a danger zone.

11. Trigger Proximity Alert:

The system activates the warning mechanism to indicate a potential risk.

12. Activate Buzzer Alarm:

A buzzer is turned on to provide an audible danger alert.

13. Send SMS Alert with GPS Location:

An alert message containing the live location link is sent to the registered mobile number.

14. Wait 30 Seconds:

The system pauses briefly before checking the conditions again to avoid repeated alerts.

15. Panic Button Monitoring:

The system continuously checks whether the panic button is pressed.

16. Immediate Emergency Mode:

Emergency actions are activated without delay.

17. Send Emergency SMS with Location:

A high-priority SMS containing the current GPS coordinates is sent instantly.

18. Activate Continuous Alerts:

The buzzer or alarm is continuously activated to indicate a critical situation.

CHAPTER 5

RESULTS AND DISCUSSIONS

The Borewell Safety Alert System was successfully designed and implemented using NodeMCU, GPS, GSM, and alert components. Multiple test scenarios were conducted to validate system performance and emergency response capability.

5.1 Results

- The system was able to accurately detect the current GPS location and continuously monitor the child's movement around the borewell.
- The geo-fencing mechanism correctly identified when the child crossed the virtual safety boundary.
- Upon entering the danger zone, the buzzer produced a loud audible alert, effectively warning nearby individuals.
- The GSM module successfully sent SMS notifications containing live GPS coordinates to the registered mobile number.
- The panic button feature worked reliably, triggering immediate emergency alerts during testing.
- Power management was stable, ensuring consistent operation through battery backup.

5.1.1 Model Performance Observation

The developed prototype model of the *Borewell Safety Alert System for Children* fig 5.1.1 functioned effectively during testing and demonstrated successful integration of all hardware modules. The NodeMCU microcontroller acted as the central control unit, continuously receiving inputs from the ultrasonic/PIR sensor embedded in the borewell model. When motion or human presence was detected, the controller immediately processed the signal and activated the alert mechanism. The GPS module provided real-time location coordinates, while the GSM/Wi-Fi communication unit transmitted an emergency alert message to the registered mobile number without manual intervention.

The model performed consistently under repeated trials, proving that the system is capable of delivering fast and accurate alerts in emergency situations. Overall, the prototype validated the concept and confirmed that the designed model can significantly reduce response time during borewell child accidents.

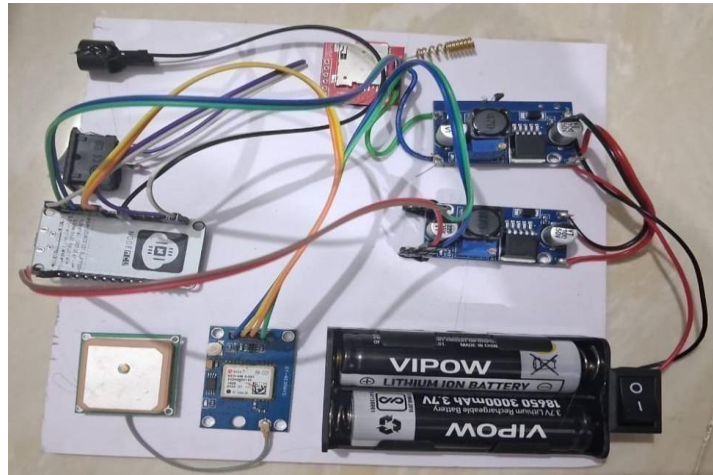


Fig5.1.1: Hardware Prototype of Borewell Safety Alert System

5.1.2 SMS-Based Location Alert Output

The image 5.1.2 shows alert messages received on a mobile phone from the Borewell Safety Alert System, indicating that a tracked user or device has entered a restricted zone near a borewell. Each message contains the distance from the hazardous point and a Google Maps link with live coordinates, demonstrating how the system can provide real-time location tracking and instant warnings. This reflects the core function of the project, where GPS-based monitoring and automated SMS alerts help inform guardians or authorities immediately, allowing quick action to prevent accidents involving children near borewells. This image represents the geographical tracking interface of our Borewell Safety Alert System, displayed through Google Maps. The blue dot indicates the current live location of the monitored child or user, while the red marker shows a targeted or restricted area where a borewell or hazardous zone is located. The map clearly outlines surrounding roads, landmarks, and buildings, which helps users navigate quickly and accurately during an emergency. This visualization confirms the effectiveness of our system in providing precise coordinates, enabling parents, guardians, or rescue teams to instantly locate a child who moves close to a borewell. By using GPS-based location identification and map integration, the system enhances real-time monitoring, supports immediate decision-making, and

reduces the delay in response time. This feature is particularly important in rural and open areas where unprotected borewells pose a high risk to children, making the system both practical and life-saving.

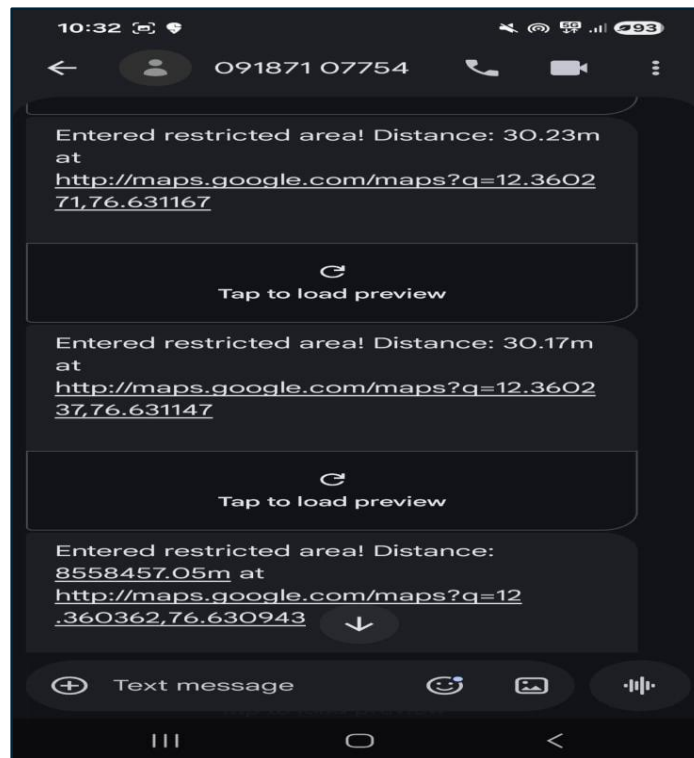
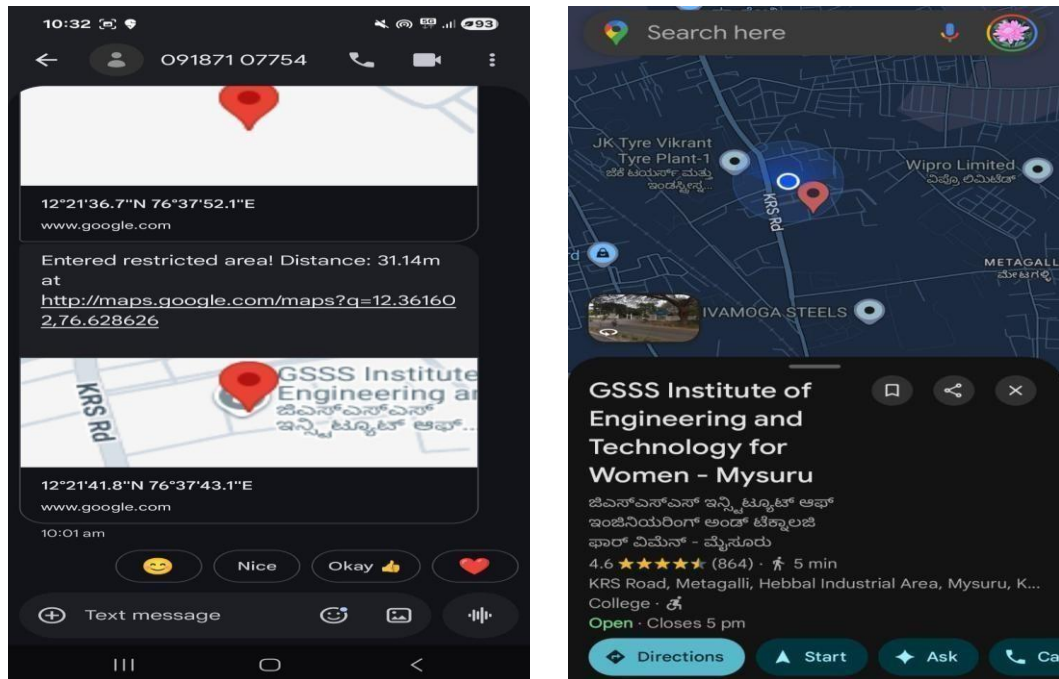


Fig5.1.2: SMS-Based Location Alert Output

5.1.3 Location Tracking Interface on Google Maps

This image 5.1.3 represents the geographical tracking interface of our Borewell Safety Alert System, displayed through Google Maps. The blue dot indicates the current live location of the monitored child or user, while the red marker shows a targeted or restricted area where a borewell or hazardous zone is located. The map clearly outlines surrounding roads, landmarks, and buildings, which helps users navigate quickly and accurately during an emergency. This visualization confirms the effectiveness of our system in providing precise coordinates, enabling parents, guardians, or rescue teams to instantly locate a child who moves close to a borewell. By using GPS-based location identification and map integration, the system enhances real-time monitoring, supports immediate decision-making, and significantly reduces the delay in response time. This feature is particularly important in

rural and open areas where unprotected borewells pose a high risk to children, making the system both practical and life-saving.



5.1.3 Google Maps Location Tracking and SMS Alert Display of Borewell

Overall, the system performed efficiently in real-time scenarios and demonstrated its capability to prevent borewell accidents.

5.2 Advantages

1. Real-time Location Tracking:

The GPS module provides continuous and accurate location monitoring at all times.

2. Automatic Geo-Fence Alerts:

The system instantly notifies when the user crosses the predefined 30-meter safety boundary.

3. Immediate Emergency Notification:

The panic button allows quick SMS alerts to be sent during critical situations.

4. Low Power and Cost-Effective:

5. Low Power and Cost-Effective: Uses NodeMCU and standard modules, making the system affordable and energy efficient

6. Fast SMS Delivery:

The GSM module sends alert messages with live GPS links within seconds.

7. Simple and Reliable Operation:

The software automatically handles GPS data, distance calculations, and alerts without human intervention.

8. Portable and Lightweight:

The compact design makes it easy to carry or install in mobile environments.

9. Accurate Distance Measurement:

TinyGPS++ ensures high-precision distance calculation for detecting geo-fence violations.

10. 24/7 Monitoring Capability:

The system can run continuously, making it suitable for security and safety applications.

11. Easy Integration and Customization:

Additional sensors or alert mechanisms can be added without major hardware changes.

5.3 Disadvantages

- **Cost and Maintenance:** High initial investment and ongoing costs for maintenance, software updates, and component replacement (e.g., batteries).
- **Infrastructure Dependency:** Requires reliable internet/cellular connectivity for timely alerts and a constant power source, which is difficult in remote areas.
- **Technical Risks:** Complex installation/operation, requiring technical expertise. System is prone to sensor failure, resulting in false alarms or critical missed alerts.

5.4 APPLICATIONS

- **Child Safety Near Borewells:**
Helps prevent accidents by sending instant alerts when a child enters the danger zone around a borewell.

- **Rural and Remote Safety Monitoring:**
Ideal for villages where many abandoned or open borewells exist and manual monitoring is difficult.
- **Emergency Rescue Support:**
The panic button and automatic GPS-based SMS help authorities respond quickly in case a child falls into a borewell.
- **Real-Time Location Tracking:**
Provides continuous GPS updates, allowing parents and rescue teams to track the exact location during emergencies.
- **Public Safety Installations:** Can be installed at construction sites, agricultural fields, and other open pits to avoid accidental falls.
- **Smart Community Alert System:** Sends SMS alerts to local panchayat members, nearby residents, or rescue teams to ensure collective response.
- **IoT-Based Safety Management:** Can be integrated with IoT dashboards for long-term monitoring and analysis of risk zones.
- **Safety for Workers:** Helps safeguard workers who operate around deep wells, tanks, or underground areas by alerting them before entering hazardous zones.

5.5 Discussions

The results confirm that the proposed safety system provides practical and timely protection against borewell-related accidents. Real-time alerts and location tracking significantly reduce rescue response time and increase the chances of saving a child.

However, the system's performance may be impacted by GPS signal quality and GSM network availability, especially in deep rural or underground areas. Despite these limitations, the project offers a low-cost, portable, and life-saving solution suitable for widespread implementation.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE OF

6.1 Conclusions

The Borewell Safety Alert System presents an effective and practical solution to address one of the most critical safety challenges faced in many rural and semi-urban regions. Accidental falls into open or abandoned borewells have become a recurring tragedy, particularly involving children. This system integrates modern technologies such as NodeMCU, GPS, GSM, geo-fencing, and an emergency panic mechanism to create a reliable safety framework. By detecting the presence of individuals near hazardous borewell zones and sending instant SMS alerts with live location coordinates, the system ensures timely intervention and significantly improves the chances of preventing accidents.

The approach is cost-effective, easy to deploy, and scalable for widespread community use. Since the design relies on commonly available components, it becomes a practical option for government bodies, local panchayats, and safety departments to implement in vulnerable areas. Moreover, the automated alert mechanism reduces dependency on manual monitoring and promotes a more proactive safety culture. With further enhancements such as cloud integration, solar power, and AI-based monitoring, the system can evolve into a fully autonomous and intelligent safety solution. Overall, this project highlights how simple yet powerful technological innovations can play a vital role in safeguarding lives and strengthening public safety infrastructure.

6.2 Future Scope

- **Advanced Connectivity and IoT Integration:**

Future systems can adopt cloud dashboards, mobile applications, and long-range communication technologies like LoRaWAN, NB-IoT, or 5G to enable centralized monitoring, real-time alerts, and seamless operation in remote rural areas.

- **AI, Automation, and Intelligent Safety Mechanisms:**

AI-based risk prediction, automated safety lids, robotic rescue mechanisms, and camerabased real-time monitoring (including night vision) can significantly enhance early detection, quick response, and on-spot preventive actions.

- **Enhanced Safety, Communication, and Government Support:** The system can be upgraded with oxygen/toxic gas sensors, two-way communication modules, and blockchain-based borewell status tracking. A government-backed centralized alert platform can ensure that authorities and emergency teams receive immediate notifications during emergencies.

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APPENDIX-A

PIN CONFIGURATIONS OF THE CONTROLLERS USED

1] ESP8266 (NodeMCU) Pin Details

The ESP8266 NodeMCU is a low-cost Wi-Fi-enabled microcontroller board that is widely used in IoT applications because of its ease of programming and integrated wireless communication. It operates on 3.3 V logic and provides multiple digital General-Purpose Input/Output (GPIO) pins that can be used for different purposes such as input, output, PWM, I2C, and interrupt handling. Although it has 17 GPIO pins, not all are recommended for regular usage because some of them also play an important role during the booting process of the microcontroller. Commonly used pins for sensors and modules include GPIO4 (D2), GPIO5 (D1), GPIO14 (D5), GPIO12 (D6), and GPIO13 (D7), whereas GPIO0 (D3), GPIO2 (D4), and GPIO15 (D8) must be used carefully since they affect the startup configuration. The ESP8266 has a single Analog-to-Digital Converter (ADC) pin, ADC0 or A0, which can read analog voltages only in the 0–1 V range, and therefore external sensors operating at higher voltages require a voltage divider before connection.

The board supports I2C communication through software, and the commonly used configuration assigns GPIO4 (D2) as SDA and GPIO5 (D1) as SCL, which makes it compatible with several I2C-based modules such as OLED displays, RTCs, and IMUs. For serial communication, the ESP8266 provides two UART interfaces; however, only UART0, consisting of TX (GPIO1) and RX (GPIO3), is fully usable for programming and serial monitoring, while UART1 supports only TX and is mainly used for debug output. The microcontroller also supports PWM signals on most GPIO pins, allowing it to control devices like LEDs and motors by varying pulse width. Unlike the ESP32, the ESP8266 does not include DAC pins for true analog output and lacks capacitive touch pins. Overall, the ESP8266 NodeMCU is an efficient and compact controller suitable for Internet of Things projects due to its built-in Wi-Fi, flexible communication support, low power consumption, and low cost.

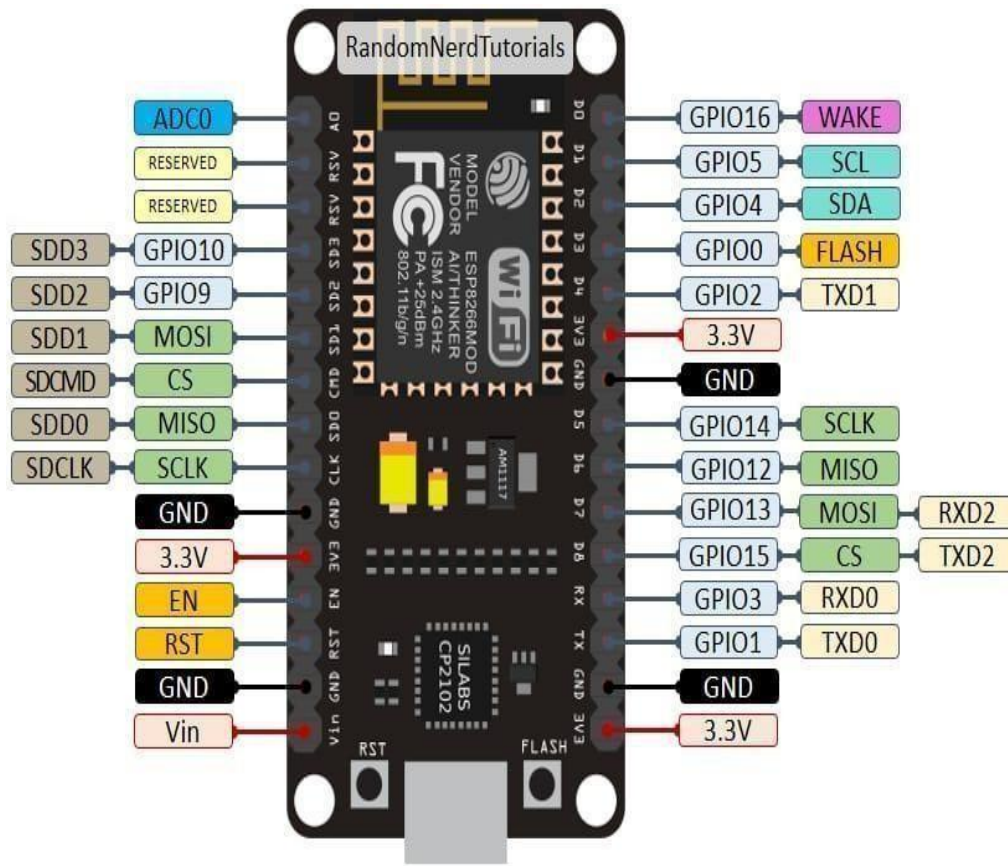


Fig : ESP8266 (NodeMCU) Pin

Power Pins of ESP8266 (NodeMCU)

The ESP8266 NodeMCU development board includes several power pins that provide the necessary operating voltages for the board and external devices. The board mainly operates on 3.3V, and therefore it has a dedicated 3V3 pin that supplies regulated 3.3V output to power sensors and modules requiring 3.3V. To power the board, the Vin pin or VU (5V) pin can be used to provide an input voltage between 5V and 12V, which is internally regulated down to 3.3V for the microcontroller. The board also has a GND pin that acts as the reference ground and must be connected to the common ground of all components in the circuit to ensure stable operation. It is important to note that powering external devices directly from 3.3V should be done carefully, as the current output capacity of the onboard regulator is limited. Supplying a voltage higher than 3.3V directly to the 3V3 pin can damage the board. Overall, the power pins of the ESP8266 play a crucial role in ensuring proper power distribution and safe operation of the NodeMCU and all the connected peripherals.

General Purpose Input Output (GPIO) Pins of ESP8266

GPIO No.	NodeMCU Pin Name
GPIO0	D3
GPIO1	TX
GPIO2	D4
GPIO3	RX
GPIO4	D2
GPIO5	D1
GPIO12	D6
GPIO13	D7
GPIO14	D5
GPIO15	D8
GPIO16	D0

Tabel 1.1: General Purpose Input Output (GPIO) Pins of ESP8266

Analog Pin of ESP8266

The ESP8266 NodeMCU has only one analog input pin, known as ADC0 or A0, which is used to measure analog signals. This pin is connected to the internal 10-bit Analog-to-Digital Converter (ADC) of the ESP8266, which converts the input analog voltage into a digital value ranging from 0 to 1023. Unlike many other microcontrollers that allow analog input up to 3.3V or 5V, the analog pin on the ESP8266 supports a maximum input voltage of 0–1V only. Therefore, when interfacing sensors that provide higher output voltages (e.g., 3.3V or 5V LDR or temperature sensors), a voltage divider or signal conditioning circuit must be used to reduce the voltage to a safe range before connecting to A0. The analog pin is generally used for reading sensor values such as light intensity, temperature variation, potentiometer position, soil moisture level, and other analog physical parameters. Since the ESP8266 has only one analog channel, it cannot directly read multiple analog sensors at the same time without using an external analog multiplexer. Overall, the A0 pin plays a crucial

role in enabling the ESP8266 to interface with analog-based sensors, making the board suitable for real-world monitoring and IoT applications.

ADC Pin	NodeMCU Pin
ADC0	A0

Tabel 1.2: Analog Pin of ESP8266

2C Interface Pins of ESP8266

The ESP8266 NodeMCU supports the I2C (Inter-Integrated Circuit) communication protocol, which allows multiple digital sensors and modules to communicate with the microcontroller using only two wires. I2C communication uses a master–slave architecture, where the ESP8266 typically acts as the master and connected sensors or modules operate as slaves. The two pins used for I2C communication are SDA (Serial Data Line) and SCL (Serial Clock Line). Unlike some microcontrollers where I2C pins are fixed in hardware, the ESP8266 uses a software- implemented I2C (Soft I2C), meaning any GPIO pin can be configured for I2C operation. However, the most commonly used and recommended pins are GPIO4 (D2) as SDA and GPIO5 (D1) as SCL, as this assignment is widely supported by libraries and ensures reliable communication. I2C is particularly useful for connecting devices such as OLED displays, RTC modules (DS1307/DS3231), temperature and humidity sensors, EEPROMs, and IMU sensors like MPU6050 because it supports multiple devices on the same bus by assigning each device a unique 7-bit or 10-bit address. Pull-up resistors are generally required on both SDA and SCL lines to maintain proper logic levels, although many breakout modules have inbuilt pull-ups. Due to its ability to connect several peripherals with minimal wiring, the I2C interface greatly enhances the flexibility and expandability of ESP8266-based IoT and embedded projects.

I2C Interface Pins of ESP8266

I2C Function	ESP8266 Pin
SCL	D1 (GPIO5)
SDA	D2 (GPIO4)

Tabel 1.3: I2C Interface Pins of ESP8266

2] GSM MODULE PIN DETAILS :

The SIM800L GSM module consists of several pins that enable power supply, communication, audio interfacing, and status monitoring. The VCC and GND pins provide the module's power input, requiring a stable voltage between 3.7V and 4.4V for reliable operation. Communication between the module and a microcontroller is achieved through the TXD and RXD pins, which handle serial data transmission and reception using AT commands. .

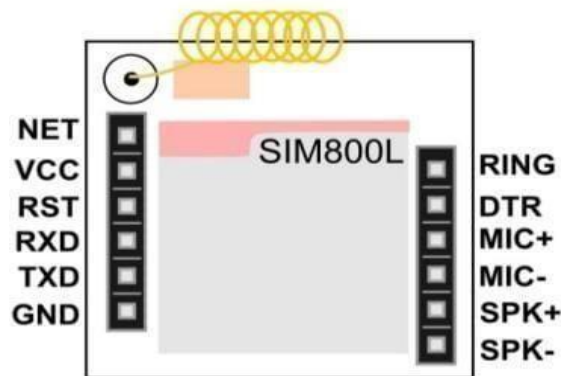


Fig 1.4: Major Part of SIM800L GSM Module with adaptor

The RST pin allows hardware resetting of the module, while the NET pin provides network status indication through an onboard LED. Additional functional pins include the RING pin, which signals incoming calls or messages, and the DTR pin, which is used to control the module's sleep mode for power saving. For audio applications, the SIM800L provides differential microphone inputs (MIC+ and MIC-) and speaker outputs (SPK+ and SPK-), enabling direct connection of external audio devices. Overall, the pin configuration of the SIM800L ensures seamless integration in GSM-based communication systems. the SIM800L module includes power-control pins such as PWRKEY, which is used to turn the module ON or OFF through an external trigger. The module also supports serial communication through its TXD and RXD pins, ensuring reliable data exchange with microcontrollers like NodeMCU or Arduino. Its carefully designed RF antenna interface provides stable signal reception, making it suitable for SMS, voice calls, and GPRS data transmission.

With all these pin functions working together, the SIM800L becomes a compact yet powerful solution for real-time communication in embedded and IoT applications.

NAME	DESCRIPTION
GND	Ground
TXD	Transmit Data
RXD	Receive Data
RST	SIM Reset
VCC	4.0V Input Set Supply
SPK-	Differential audio output(SPEAKER IN)
SPK+	Differential audio output(SPEAKER OUTPUT)
MIC-	Differential audio output(MICROPHONE N)
MIC+	Differential audio input(MICROPHONE P)
DTR	Data Terminal Ready
RING	Ring Indicator

Fig 1.5: SIM800L GSM Configuration

